

CS 660: Data Mining For Decision Making

Lecture 5 (Week 2)

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Scaling the Heights

Numeric Attributes

Table 4.4 Numeric Weather Data with Summary Statistics

Outlook			Temperature			Humidity			Windy			Play	
	yes	no		yes	no		yes	no		yes	no	yes	no
sunny	2	3		83	85		86	85	false	6	2	9	5
overcast	4	0		70	80		96	90	true	3	3		
rainy	3	2		68	65		80	70					
				64	72		65	95					
				69	71		70	91					
				75			80						
				75			70						
				72			90						
				81			75						
sunny	2/9	3/5	<i>mean</i>	73	74.6	<i>mean</i>	79.1	86.2	false	6/9	2/5	9/14	5/14
overcast	4/9	0/5	<i>std. dev.</i>	6.2	7.9	<i>std. dev.</i>	10.2	9.7	true	3/9	3/5		
rainy	3/9	2/5											

Handling Numeric Attributes in Naïve Bayes

The probability density function for a normal distribution with mean μ and standard deviation σ is given by the rather formidable expression

$$f(x) = \frac{1}{\sqrt{2\pi}\sigma} e^{-\frac{(x-\mu)^2}{2\sigma^2}}$$

But fear not! All this means is that if we are considering a *yes* outcome when *temperature* has a value of, say, 66, we just need to plug $x = 66$, $\mu = 73$, and $\sigma = 6.2$ into the formula. So the value of the probability density function is

$$f(\text{temperature} = 66 \mid \text{yes}) = \frac{1}{\sqrt{2\pi} \times 6.2} e^{-\frac{(66-73)^2}{2 \times 6.2^2}} = 0.0340$$

And by the same token, the probability density of a *yes* outcome when *humidity* has a value of, say, 90, is calculated in the same way:

$$f(\text{humidity} = 90 \mid \text{yes}) = 0.0221$$

Handling Numeric Attributes in Naïve Bayes

Table 4.5 Another New Day

Outlook	Temperature	Humidity	Windy	Play
Sunny	66	90	true	?

Likelihood of *yes* = $2/9 \times 0.0340 \times 0.0221 \times 3/9 \times 9/14 = 0.000036$

Likelihood of *no* = $3/5 \times 0.0279 \times 0.0381 \times 3/5 \times 5/14 = 0.000137$

Missing Values

Table 4.3 A New Day

Outlook	Temperature	Humidity	Windy	Play
Sunny	cool	high	true	?

One of the really nice things about Naïve Bayes is that missing values are no problem at all. For example, if the value of *outlook* were missing in the example of Table 4.3, the calculation would simply omit this attribute, yielding

$$\text{Likelihood of } yes = 3/9 \times 3/9 \times 3/9 \times 9/14 = 0.0238$$

$$\text{Likelihood of } no = 1/5 \times 4/5 \times 3/5 \times 5/14 = 0.0343$$

These two numbers are individually a lot higher than they were before because one of the fractions is missing. But that's not a problem because a fraction is missing in both cases, and these likelihoods are subject to a further normalization process. This yields probabilities for *yes* and *no* of 41% and 59%, respectively.

Full dataset

Table 4.2 Weather Data with Counts and Probabilities

Outlook			Temperature			Humidity			Windy			Play	
	yes	no		yes	no		yes	no		yes	no	yes	no
sunny	2	3	hot	2	2	high	3	4	false	6	2	9	5
overcast	4	0	mild	4	2	normal	6	1	true	3	3		
rainy	3	2	cool	3	1								
sunny	2/9	3/5	hot	2/9	2/5	high	3/9	4/5	false	6/9	2/5	9/14	5/14
overcast	4/9	0/5	mild	4/9	2/5	normal	6/9	1/5	true	3/9	3/5		
rainy	3/9	2/5	cool	3/9	1/5								

Activity

- Read Naïve Bayes from Witten's Book

Thank you!

Comments and Questions most welcome!